Mingda Liu

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Education

National University of Singapore *August 2026 - December 2027 (Expected)*
Master of Computing in Artificial Intelligence

Institute of Science Tokyo *April 2022 - March 2026*
Formerly Tokyo Institute of Technology (QS World University Rankings 2026: #85)
B.S. in Mathematical and Computing Science
*GPA: **94.59/100*** *Rank: **1/51***

Professional Experience

Researcher (Full-time)**, Institute of Science Tokyo** *April 2026 - July 2026*
Fujisawa Laboratory

Focused on **LLM/VLM post-training and Agentic RL**, with research aimed at improving inference efficiency, building scalable data flywheels, enabling reliable agent self-improvement, and optimizing retrieval for knowledge-intensive reasoning.

Selected Projects

RECAST: Modularizing Real-World Repositories into Composable Training Environments for Coding Agents
Coding Agents · Data Synthesis · RL Environments

Developed a **framework** that turns existing code repositories into reusable training assets, allowing each newly collected codebase to generate many verifiable coding tasks and compound its value in a scalable coding-agent data flywheel.

- Recursively decomposed repositories into **dependency-aware functional modules** and recombined them into controllable start/goal states, transforming static code into a combinatorial task space instead of one-off issue/patch pairs.
- Multiplied the training value of existing code through **modular recomposition**—for example, 7 functional capabilities identified and peeled from a single repository were recombined into **257 distinct, verified training tasks**.
- Preserved real source code as ground-truth targets through **deletion-only synthesis**, with sandboxed execution, replayable provenance, blind mutation probes, and persistent counterexamples to harden RL rewards against weak tests and reward hacking.

ToolBranch: Efficient Adaptive Tool Use for LLM/VLM Agents
LLM/VLM Post-Training · Agentic RL · Tool Use

Developed an **on-policy RL training method** that teaches LLM/VLM agents when to solve tasks directly versus use external tools, improving task quality while avoiding unnecessary tool calls and inference overhead.

- Introduced **Scaffolded Branch-GRPO**, explicitly generating paired tool-use and no-tool on-policy branches for each training sample so both decisions retain comparative reward signals throughout training, preventing the exploration collapse of vanilla GRPO.
- Across **When2Tool, V*, MathVista, MMBench, MME, MMSI-Bench, and SPAR-Bench**, improved aggregate accuracy by **8.87 pp** over vanilla GRPO while reducing tool usage by **40.73%** and token consumption by **29.42%**.
- Built the end-to-end **Agentic RL training and evaluation stack** for **multimodal LLMs (e.g., Qwen3-VL)** with **verl/VerlTool, vLLM, Ray**, supporting distributed training, controlled ablations, and branch-level diagnostics.

Publications

Norm Anchors Make Model Edits Last ***NeurIPS*** *2026*
Mingda Liu[*], Zhenghan Zhu[*], Ze'an Miao[*], Katsuki Fujisawa.
LLM Post-Training · Lifelong Learning. Proposed Norm Anchors, a lightweight method that extends how many edits a model can sustain by more than 4× at negligible cost, making long-term model maintenance more reliable.

cMoLLM at Scale: Horizontal Scaling Laws for Convolutionally-Gated Mixture-of-LLMs ***ICML*** *2026*
Xin Yang, Yemin Wang, **Mingda Liu**, Letian Li, Shuaishuai Cao, Zhengxiao He, Ryan Dong.
LLM Architecture · Efficient Scaling. Developed cMoLLM to scale model capacity through dynamically routed end-to-end model streams, improving model quality under matched compute without proportionally increasing computation.

Fork of Thought: Eliciting Self-Triggered Rethinking in LLM Agents via Branch-Sampled Reinforcement Learning *Under review,* ***ICLR*** *2027*
Junjie Meng[*], Wei Wu[*], **Mingda Liu**[*], Wen-Jie Shu, Xiaoru Liu, Jiqcheng Lu, Jun Liu, Hui Xiong.
Agentic RL · Self-Evolution. Trained long-horizon agents to selectively invoke fresh-context verification and recover from earlier mistakes in-flight, enabling more reliable multi-step automation without relying on a separate stronger critic.

BRIDGE: Bilevel Retrieval Credit-Aware Agentic Reinforcement Learning *Under review,* ***ICLR*** *2027*
Quan Xiao[*], **Mingda Liu**[*], Gaowen Liu, Katsuki Fujisawa, Tianyi Chen.
Agentic RL · Retrieval. Closed the information-credit gap between search agents and retrievers, improving knowledge-intensive reasoning while outperforming a joint-training baseline at roughly half the training time.
[*]*Equal contribution.*

Honors & Scholarships

Undergraduate Valedictorian, Institute of Science Tokyo Degree Conferment Ceremony *2026*
Outstanding International Undergraduate Scholarship (Best Award), Institute of Science Tokyo *2022 - 2026*
Fuji Seal Foundation Scholarship (Grant-type), Fuji Seal Foundation *2024*
Monbukagakusho Honors Scholarship for Privately-Financed International Students (JASSO) *2022*

Skills

Programming	Python, C++, Go, C, R, Java
ML / Agent RL	PyTorch, verl, VerlTool, Hugging Face
Distributed / Inference	vLLM, Ray, FSDP
Retrieval / Experimentation	FAISS, W&B
Systems	Linux, multi-node GPU training, SGE/cluster workflows
Languages	Chinese (Native), Japanese (Business), English (TOEFL iBT 101)

Additional Experience

Teaching Assistant / Instructor (4 years total) *2022 - 2026*
Taught and mentored **100+ students** in programming, mathematics, and physics; created assignments and provided office-hours and debugging support.

Competitive Programming: AtCoder, top **6%** globally.